Practical 3: Big Query

1.

Q1	WHERE Clause

Filter all transactions that occurred in the year 2023.

Expected Output: *All columns*

```
--Q1 WHERE clause
SELECT `Transaction ID`,
    Date,
    `Customer ID`,
    Gender,
    Age,
    `Product Category`,
    Quantity,
    `Price per Unit`,
    `Total Amount`,
FROM `project-3d798d88-9e9a-4860-93a.Practice_data.Retail sales dataset`
WHERE EXTRACT(YEAR FROM Date)= 2023
```

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	191	2023-10-18	CUST191	Male	64	Beauty	1	25	25
2	204	2023-09-28	CUST204	Male	39	Beauty	1	25	25
3	230	2023-04-23	CUST230	Male	54	Beauty	1	25	25
4	232	2023-02-06	CUST232	Female	43	Beauty	1	25	25
5	309	2023-12-23	CUST309	Female	26	Beauty	1	25	25
6	310	2023-10-12	CUST310	Female	28	Beauty	1	25	25
7	363	2023-06-03	CUST363	Male	64	Beauty	1	25	25
8	371	2023-02-21	CUST371	Female	20	Beauty	1	25	25
9	397	2023-03-10	CUST397	Female	30	Beauty	1	25	25
10	454	2023-02-22	CUST454	Female	46	Beauty	1	25	25

2.

Q2	Filtering + Conditions

Display all transactions where the Total Amount is more than the average Total Amount of the entire dataset.

Expected Output: *All columns*

```
--Q2
SELECT *
FROM `project-3d798d88-9e9a-4860-93a.Practice_data.Retail sales dataset`
WHERE `Total Amount`> (SELECT AVG(`Total Amount`) FROM `project-3d798d88-9e9a-4860-93a.Practice_data.Retail sales dataset`)
```

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	21	2023-01-14	CUST021	Female	50	Beauty	1	500	500
2	28	2023-04-23	CUST028	Female	43	Beauty	1	500	500
3	128	2023-07-05	CUST128	Male	25	Beauty	1	500	500
4	220	2023-03-03	CUST220	Male	64	Beauty	1	500	500
5	238	2023-01-17	CUST238	Female	39	Beauty	1	500	500
6	364	2023-08-23	CUST364	Female	19	Beauty	1	500	500
7	408	2023-04-15	CUST408	Female	64	Beauty	1	500	500
8	537	2023-06-03	CUST537	Female	21	Beauty	1	500	500
9	539	2023-06-08	CUST539	Male	25	Beauty	1	500	500
10	541	2023-07-29	CUST541	Male	56	Beauty	1	500	500

3.

Q3 Aggregate Functions

Calculate the total revenue (sum of Total Amount).

Expected Output: *Total_Revenue*

```
--Q3
SELECT SUM(`Total Amount`) AS Total_Revenue
FROM `project-3d798d88-9e9a-4860-93a.Practice_data.Retail sales dataset`
```

Row	Total_Revenue
1	456000

4.

Q4 DISTINCT

Display all distinct Product Categories in the dataset.

Expected Output: *Product_Category*

```
--Q4
SELECT DISTINCT(`Product Category`)
FROM `project-3d798d88-9e9a-4860-93a.Practice_data.Retail sales dataset`
```

Row	Product Category
1	Beauty
2	Clothing
3	Electronics

5.

Q5	GROUP BY

For each Product Category, calculate the total quantity sold.

Expected Output: *Product_Category, Total_Quantity*

```
--Q5
SELECT `Product Category`,
SUM(Quantity) AS Total_quantity
FROM `project-3d798d88-9e9a-4860-93a.Practice_data.Retail sales dataset`
GROUP BY `Product Category`
```

Row	Product Category	Total_quantity
1	Beauty	771
2	Clothing	894
3	Electronics	849

6.

Q6	CASE Statement

Create a column called Age_Group that classifies customers as:

- 'Youth' (<30)

- 'Adult' (30–59)
- 'Senior' (60+)

Expected Output: *Customer_ID, Age, Age_Group*

```sql
--Q6
SELECT`Customer ID`,
       Age,
CASE
    WHEN Age<30 THEN 'Youth'
    WHEN Age BETWEEN 30 AND 59 THEN 'Adult'
    WHEN Age>=60 THEN 'Senior'
END AS Age_group
FROM `project-3d798d88-9e9a-4860-93a.Practice_data.Retail sales dataset`
```

Row	Customer ID	Age	Age_group
22	CUST344	42	Adult
23	CUST376	64	Senior
24	CUST387	44	Adult
25	CUST396	55	Adult
26	CUST670	27	Youth
27	CUST733	34	Adult
28	CUST747	23	Youth
29	CUST749	42	Adult
30	CUST793	54	Adult
31	CUST802	46	Adult

7.

Q7 Conditional Aggregation

For each Gender, count how many high-value transactions occurred (where Total Amount > 500).

Expected Output: *Gender, High_Value_Transactions*

```sql
SELECT Gender,
       COUNT(CASE WHEN `Total Amount`>500 THEN 1 END) AS High_Value_Transactions
FROM `project-3d798d88-9e9a-4860-93a.Practice_data.Retail sales dataset`
GROUP BY Gender;
```

Row	Gender	High_Value_Tran...
1	Male	144
2	Female	155

8.

Q8	HAVING Clause

For each Product Category, show only those categories where the total revenue exceeds 5,000.

Expected Output: *Product_Category, Total_Revenue*

```sql
--Q8
SELECT `Product Category`,
SUM(`Total Amount`) AS Total_Revenue
FROM `project-3d798d88-9e9a-4860-93a.Practice_data.Retail sales dataset`
GROUP BY `Product Category`
HAVING SUM(`Total Amount`)> 5000
```

Row	Product Category	Total_Revenue
1	Beauty	143515
2	Clothing	155580
3	Electronics	156905

9.

Q9	Calculated Fields

Display a new column called Unit_Cost_Category that labels a transaction as:

- 'Cheap' if Price per Unit < 50
- 'Moderate' if Price per Unit between 50 and 200
- 'Expensive' if Price per Unit > 200

Expected Output: *Transaction_ID, Price_per_Unit, Unit_Cost_Category*

```sql
--Q9
SELECT`Transaction ID`,
      `Price per Unit`,
CASE
    WHEN `Price per Unit`<30 then 'Cheap'
    WHEN `Price per Unit` BETWEEN 50 AND 200 THEN 'Moderate'
    WHEN `Price per Unit`>200 THEN 'Expensive'
END AS Unit_Cost_Category
FROM `project-3d798d88-9e9a-4860-93a.Practice_data.Retail sales dataset`
```

Row	Transaction ID	Price per Unit	Unit_Cost_Category
1	191	25	Cheap
2	204	25	Cheap
3	230	25	Cheap
4	232	25	Cheap
5	309	25	Cheap
6	310	25	Cheap
7	363	25	Cheap
8	371	25	Cheap
9	397	25	Cheap
10	454	25	Cheap

10.

Q10 Combining WHERE + CASE

Display all transactions from customers aged 40 or older and add a column Spending_Level showing:

- 'High' if Total Amount > 1000
- 'Low' otherwise

Expected Output: *Customer_ID, Age, Total_Amount, Spending_Level*

```sql
--10
SELECT`Customer ID`,
       Age,
       `Total Amount`,
CASE
   WHEN `Total Amount`> 1000 THEN 'High'
   ELSE 'Low'
END AS Spending_Level
FROM `project-3d798d88-9e9a-4860-93a.Practice_data.Retail sales dataset`
WHERE Age>=40
```

Row	Customer ID	Age	Total Amount	Spending_Level
151	CUST522	46	1500	High
152	CUST611	51	1500	High
153	CUST693	41	1500	High
154	CUST720	56	1500	High
155	CUST480	42	2000	High
156	CUST503	45	2000	High
157	CUST592	46	2000	High
158	CUST832	47	2000	High
159	CUST103	59	25	Low
160	CUST205	43	25	Low
161	CUST206	61	25	Low